

Homework 1

Give exact answers and show your reasoning. Do not use calculators, graphing software, or other electronic tools. All functions are real-valued, and all angles are in radians. Use the conventions for inverse trigonometric functions in Lecture 2; in particular, arcsec has range $[0, \pi] \setminus \{\pi/2\}$.

1. Find the supremum of each set and state whether the supremum belongs to the set. Give a brief explanation.

(a) $A = \{x \in \mathbb{R} : x^2 < 3\}$.

(b) $B = \{q \in \mathbb{Q} : q < 5/2\}$.

(c) $C = [-1, 2] \cup \{4\}$.

2. Evaluate:

(a) $\left(\frac{81}{16}\right)^{-3/4}$, (b) $32^{2/5}$, (c) $\left(\sqrt[3]{25}\right)^{3/2}$.

3. Simplify the following expressions, assuming $a, x, y > 0$:

(a) $\frac{(a^{\sqrt{5}-1})^{\sqrt{5}+1}}{a^2}$, (b) $\frac{(x^{3/2}y^{-1/2})^2}{xy^{-2}}$.

4. Let $a, b > 0$ and let n be a positive integer. Use the definition of the positive n th root to prove

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}.$$

Do not use the laws of rational exponents in your proof.

5. Solve the inequality

$$\left(\frac{1}{2}\right)^{x^2-x} \geq \frac{1}{4}.$$

Give your answer as an interval.

6. Solve $4^x - 5 \cdot 2^x + 4 = 0$ for $x \in \mathbb{R}$.

7. Let $f(x) = \sqrt{x+2}$ and $g(x) = \sqrt{3-x}$, each on its largest real domain. Find a formula and the domain for each function:

(a) $f + g$, (b) fg , (c) $-2f$, (d) f/g .

8. Let $f(x) = \sqrt{x}$ and $g(x) = 1/(x - 2)$, each on its largest real domain. Find $f \circ g$ and $g \circ f$, including their domains. Are these the same function?

9. Let

$$F(x) = \frac{x^2 - 9}{x - 3}, \quad x \in \mathbb{R} \setminus \{3\}, \quad G(x) = x + 3, \quad x \in \mathbb{R}.$$

Are F and G equal as functions? Explain. State a restriction of the domain of G that makes it equal to F .

10. Find the inverse of $f(x) = (x + 2)^2 - 3$ with domain $(-\infty, -2]$. State the domain and range of the inverse, and verify both inverse identities on the appropriate domains.

11. Let $f(x) = (x + 1)/(x - 1)$, with domain $\mathbb{R} \setminus \{1\}$.

(a) Show that $f(x) \neq 1$ for every x in its domain, and compute $f(f(x))$.

(b) Deduce the range of f , show that it is one-to-one, and find f^{-1} . Is f the identity function?

12. Let $h(x) = 2^{x+1} - 2$, with domain $\mathbb{R}$.

(a) Find h^{-1} , including its domain and range.

(b) Sketch h , h^{-1} , and $y = x$ on the same axes by hand. Label three points on each function, their intercepts, and their asymptotes.

13. Evaluate:

$$(a) \log_9 27, \quad (b) \log_{1/4} 8, \quad (c) \ln(e^3 \sqrt{e}), \quad (d) e^{\ln 7 - 2 \ln \sqrt{7}}.$$

14. Find the domain of

$$L(x) = \ln \left(\frac{(x - 1)^2}{x + 2} \right).$$

Expand $L(x)$ using logarithm laws so that your expression is valid on its entire domain. Explain why $2 \ln(x - 1) - \ln(x + 2)$ does not suffice.

15. Solve each equation, checking the domain of the original expression:

$$(a) \ln(x+2) + \ln(x-1) = \ln 4, \quad (b) 2 \ln x = \ln(3x+4).$$

16. Solve $\log_{1/2}(x-1) \geq -2$. Explain why the direction of the inequality changes when the logarithm is removed.

17. Let $a, b > 0$, with $a, b \neq 1$.

$$(a) \text{ Prove } \log_a b = \frac{1}{\log_b a}.$$

$$(b) \text{ If also } u, v > 0 \text{ and } v \neq 1, \text{ prove } \frac{\log_a u}{\log_a v} = \log_v u.$$

18. Evaluate:

$$(a) \arcsin(-\sqrt{3}/2), \quad (b) \arccos(-1/2), \quad (c) \arctan 1, \quad (d) \operatorname{arcsec}(-2).$$

19. Evaluate each composition. State the range restriction that determines the answer.

$$(a) \arcsin(\sin(7\pi/6)).$$

$$(b) \arccos(\cos(7\pi/4)).$$

$$(c) \arctan(\tan(5\pi/6)).$$

$$(d) \operatorname{arcsec}(\sec(5\pi/4)).$$

20. Express each function using only algebraic operations and square roots, and give its domain. Justify the signs of the square roots.

$$(a) \sin(\arctan x), \quad (b) \tan(\arccos x).$$